

Technical Deep-Dive: PCA and GMM in GIV

Intuition, Mathematics, and Implementation

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Abstract

This document provides a comprehensive technical overview of Principal Component Analysis (PCA) and the Generalized Method of Moments (GMM), focusing on their application within the Granular Instrumental Variables (GIV) framework (Gabaix and Koijen, 2022). We bridge the gap between general statistical theory and practical granular identification.

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1 The Core Challenge: Latent Factor Loadings

In the simplest GIV model, we assume all entities i have identical sensitivity to macro shocks η_t . However, if large entities are systematically more sensitive to aggregate shocks than small ones, the simple GIV instrument $z_t = y_{St} - y_{Et}$ becomes *leaky*—it inadvertently instruments with aggregate shocks.

To fix this, we need the **Optimal Weights** Γ :

$$\Gamma = S - \lambda(\lambda'\lambda)^{-1}\lambda'S \quad (1)$$

The challenge is that the factor loadings λ_i are typically **unobserved (latent)**. We use PCA and GMM to recover them.

2 Principal Component Analysis (PCA)

2.1 General Theory and Intuition

Intuition: PCA is a dimension-reduction technique. Imagine a swarm of data points in N dimensions (e.g., N firms' returns over time). PCA finds the "axis of maximum variance"—the direction in which the data stretches the most. This first axis represents the dominant common factor (e.g., the "market" or "macroeconomy").

Mathematical Setup: Given a $T \times N$ matrix Y , we seek a linear combination of columns Yw that maximizes variance:

$$\max_w \text{Var}(Yw) \quad \text{s.t.} \quad \|w\| = 1 \quad (2)$$

This is equivalent to finding the eigenvectors of the covariance matrix $\Sigma = \frac{1}{T}Y'Y$. The first eigenvector (associated with the largest eigenvalue) provides the weights for the first Principal Component (PC1).

2.2 PCA in the Context of GIV

In GIV, we assume the outcomes y_{it} follow a factor structure:

$$y_{it} = \phi x_{it} + \lambda_i \eta_t + u_{it} \quad (3)$$

If the idiosyncratic shocks u_{it} are small relative to the macro shock η_t , then PCA on the matrix of outcomes Y will "pull out" the latent macro shock $\hat{\eta}_t$ as the first principal component.

Practical Steps:

1. **De-mean:** Center each firm's time series.
2. **Factor Extraction:** Apply PCA to the $T \times N$ outcome matrix.
3. **Loadings:** The coefficients of firm i on the first PC are the estimated loadings $\hat{\lambda}_i$.
4. **Orthogonalization:** Use $\hat{\lambda}_i$ to calculate Γ_i .

3 Generalized Method of Moments (GMM)

3.1 General Theory and Intuition

Intuition: OLS minimizes residuals. GMM "matches moments." If a theory says "the average error should be zero" and "the error should be uncorrelated with Z ", GMM finds the parameters that make these sample moments as close to zero as possible.

Mathematical Setup: Define a vector of moment conditions $g(\theta)$:

$$E[g(y_t, x_t, \theta)] = 0 \quad (4)$$

The GMM estimator $\hat{\theta}$ minimizes the quadratic form:

$$J(\theta) = \left(\frac{1}{T} \sum g_t(\theta) \right)' W \left(\frac{1}{T} \sum g_t(\theta) \right) \quad (5)$$

where W is a weighting matrix.

3.2 GMM in the Context of GIV

GMM is more robust than PCA because it handles the endogeneity of x_{it} and the factor structure simultaneously. Gabaix and Koijen (2022) use an **Iterative GMM** approach.

The Moments:

1. $E[(y_{it} - \psi x_{it} - \lambda_i \eta_t) z_t] = 0$ (Instrument validity)
2. $E[(y_{it} - \psi x_{it} - \lambda_i \eta_t) \mathbf{1}] = 0$ (Unbiasedness)

Practical Steps (Iterative Algorithm):

1. Start with $\psi^{(0)}$ (e.g., from OLS).
2. Calculate residuals: $e_{it} = y_{it} - \psi^{(0)} x_{it}$.
3. Estimate λ_i from e_{it} using PCA.
4. Construct the GIV z_t using these λ_i .
5. Re-estimate $\psi^{(1)}$ via IV using z_t .
6. Repeat until ψ and λ converge.

4 Sample Data and Code

4.1 Data Generation (Intuition)

We simulate a granular economy where:

- $N = 5$ firms, with sizes $S = [0.5, 0.2, 0.15, 0.1, 0.05]$.
- Large firms are **more sensitive** to the macro shock ($\lambda_1 = 2.0$, while $\lambda_5 = 0.5$).
- This creates a "leaky" simple GIV.

4.2 Python Implementation

```
import numpy as np
import pandas as pd
from sklearn.decomposition import PCA

# 1. Generate Sample Data
T, N = 100, 5
S = np.array([0.5, 0.2, 0.15, 0.1, 0.05])
lambdas = np.array([2.0, 1.5, 1.0, 0.8, 0.5]) # Large firms more
sensitive

eta = np.random.normal(0, 1, T) # Macro shock
u = np.random.normal(0, 1, (T, N)) # Idiosyncratic shocks
Y = np.outer(eta, lambdas) + u # Outcomes

# 2. PCA Approach to estimate Lambda
pca = PCA(n_components=1)
pca.fit(Y)
lambda_hat = pca.components_[0] # Estimated loadings

# 3. Construct Gamma (Optimal Weights)
# Project S orthogonal to lambda_hat and a constant (1)
X = np.column_stack([np.ones(N), lambda_hat])
Gamma = S - X @ np.linalg.inv(X.T @ X) @ X.T @ S

# 4. Construct the GIV
z_t = Y @ Gamma

print("First 5 GIV realizations:", z_t[:5])
```

Listing 1: GIV Implementation with PCA-estimated loadings

5 Summary for Researchers

When presenting GIV, emphasize:

- **PCA** is the tool for *discovery*: it finds the latent macro patterns that we must filter out.
- **GMM** is the tool for *consistency*: it ensures our structural parameters (like elasticities) are identified even when the factor structure is complex.
- **Validity** comes from orthogonality: if your weights Γ aren't perpendicular to λ , your instrument isn't clean.